It was clear how Newton's gravitational law explains tidal forces: They are produced by a difference in the strength and direction of gravity's pull, from one place to another. But Newton's law, with its gravitational force that depends on distance, had to be wrong; it violated the principle of relativity ("in whose frame was the distance to be measured?"). Einstein's challenge was to formulate a completely new gravitational law that is simultaneously compatible with the principle of relativity and explains tidal gravity in some new, simple, compelling way.

From mid-1911 to mid-1912, Einstein tried to explain tidal gravity by assuming that time is warped, but space is flat. This radical-sounding idea was a natural outgrowth of gravitational time dilation: The different rates of flow of time near the ceiling and the floor of a room on Earth could be thought of as a warpage of time. Perhaps, Einstein speculated, a more complicated pattern of time warpage might produce all known gravitational effects, from tidal gravity to the elliptical orbits of the planets to even the anomalous perihelion shift of Mercury.

After a twelve-month pursuit of this intriguing idea, Einstein abandoned it, and for a good reason. Time is relative. Your time is a mixture of my time and my space (if we move with respect to each other), and therefore, if your time is warped but your space is flat, then my time and my space must both be warped, as must be everybody else's. You and only you will have a flat space, so the laws of physics must be picking out your reference frame as fundamentally different from all others—in violation of the principle of relativity.

Nevertheless, time warpage "smelled right" to Einstein, so perhaps—he reasoned—everybody's time is warped and, inevitably alongside that, everybody's space is warped. Perhaps these combined warpages could explain tidal gravity.

The idea of a warpage of *both* time and space was rather daunting. Since the Universe admits an infinite number of different reference frames, each moving with a different velocity, there would have to be an infinity of warped times and an infinity of warped spaces! Fortunately, Einstein realized, Hermann Minkowski had provided a powerful tool for simplifying such complexity: "Henceforth, space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the